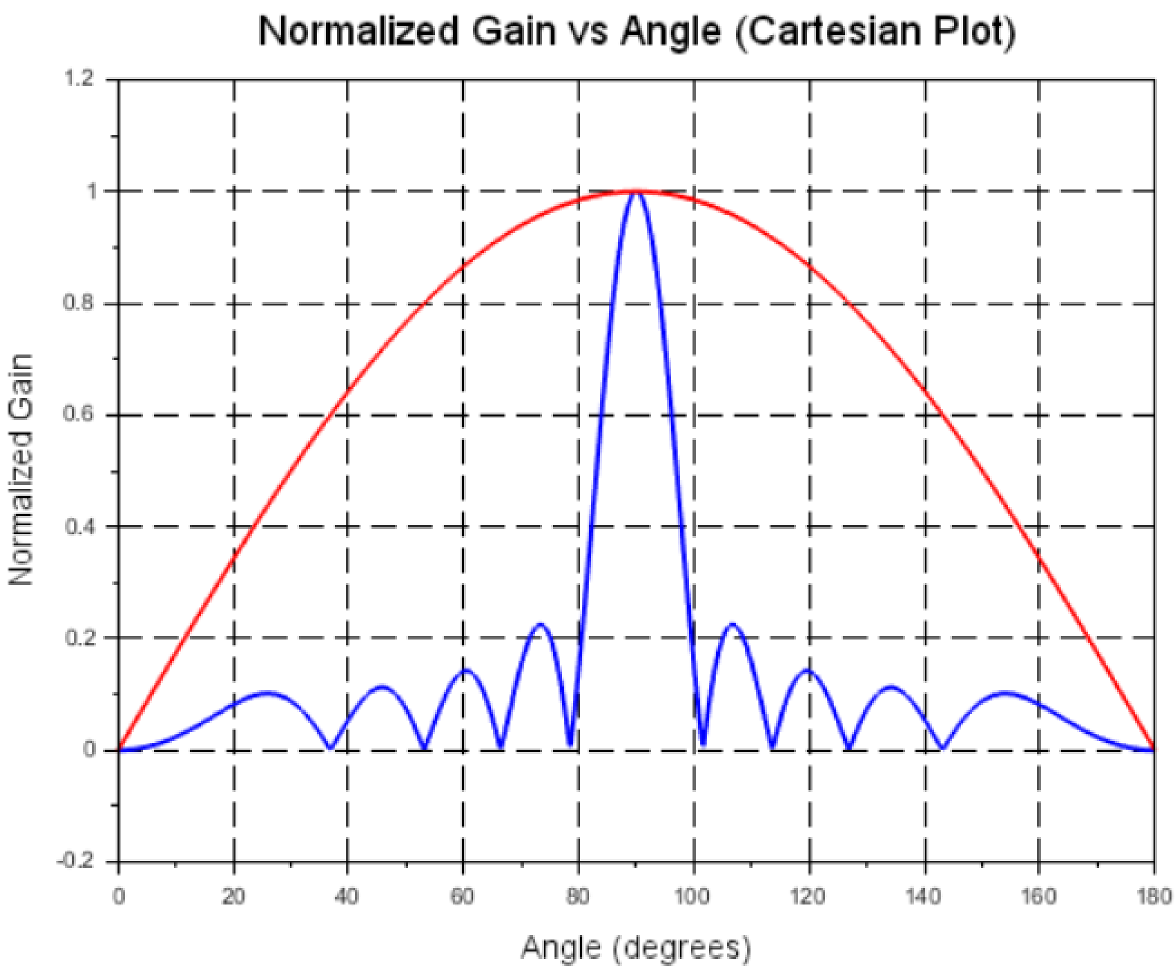
EXP:17

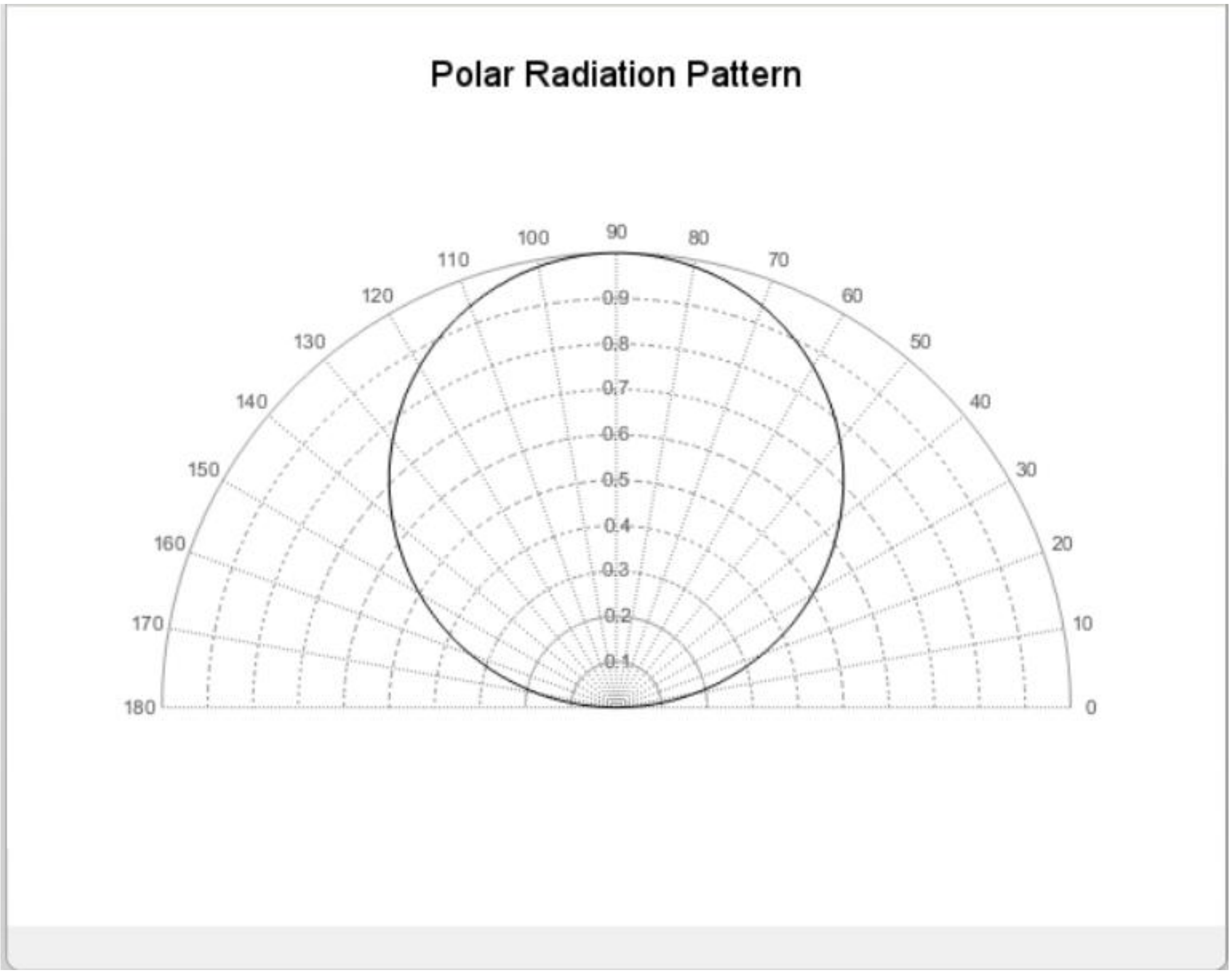
Determination and plotting of Antenna Gain Calculation.

```
clc;
clear;
N = 10;                  // Number of elements
d_lambda = 0.5;          // Distance between elements in terms of wavelength
beta = 0;                // Phase shift
theta_deg = linspace(0, 180, 1000);
theta = theta_deg * (%pi / 180); // Convert to radians
psi = 2 * %pi * d_lambda * cos(theta) + beta;
AF = zeros(1, length(psi));
for k = 1:length(psi)
    if abs(sin(psi(k)/2)) < 1e-7 then
        AF(k) = N; // Main beam peak limit
    else
        AF(k) = sin(N * psi(k) / 2) / (N * sin(psi(k) / 2));
    end
end
AF_normalized = abs(AF) / max(abs(AF));
scf(0); // Open graphic window 1
clf();
plot(theta_deg, AF_normalized, 'thickness', 2, 'color', 'blue');
title("Array Factor vs Theta", "fontsize", 4);
xlabel("Theta (degrees)", "fontsize", 3);
ylabel("Normalized Array Factor", "fontsize", 3);
ax = gca();
ax.grid = [1, 1];         // Turn on grid for x and y
ax.grid_style = [2, 2];   // Dashed style
set(ax, "data_bounds", [0, -0.05; 180, 1.05]);
scf(1); // Open graphic window 2
clf();
rho = abs(sin(theta)); // Normalized gain or pattern example
polarplot(theta, rho);
title("Polar Radiation Pattern", "fontsize", 4);
```

OUTPUT:



Normalized Gain vs Angle (Cartesian Plot)
Normalized Gain
Angle (degrees)
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0
20
40
60
80
100
120
140
160
180


Polar Radiation Pattern
90
100
80
110
70
120
60
130
50
140
40
150
30
160
20
170
10
180
0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1